NMC 2012 PRIMARY LEVEL PAPER 1 QUESTIONS

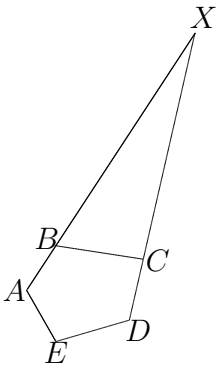
SECTION A

Qn. 1. If $\frac{x}{y} = 1$, what is the value of $x - y$?

Qn. 2. An operation is defined as $a * b = a + \frac{b}{2}$. What is the value of $-8 * 6$?

Qn. 3. Allen is paid \$51 for the first week of her job. For each week after the first, she will be paid \$100. How many weeks in total does she have to work for her average weekly pay to be \$93?

Qn. 4.



When the sides AB and DC of a regular pentagon $ABCDE$ are extended, they intersect at point X. Angle $X =$ _____ degrees.

Qn. 5. Seven consecutive positive integers/numbers have a sum of 91. What is the largest of these integers/numbers?

Qn. 6. The length of a rectangle is 3 times its width. A new rectangle is created by decreasing the length of the original rectangle by half. By what factor must the original width be multiplied, if the area remains unchanged?

Qn. 7. Kato has a square garden with an area of 1600 square metres. She plans to create a straight path from the middle of one side to the middle of the opposite side. If the path is 3 metres wide, how many square metres of garden will be left?

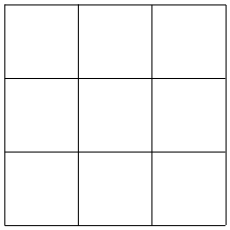
Qn. 8. A certain Primary School is to serve its 1600 pupils with 2 slices of bread at break time. The supplier of the bread has a machine that cuts each loaf of bread into 18 slices. The school never serves the pupils with the slices at both ends of each loaf. The ends are hard and as such, thrown into the dust bin. How many loaves of bread must the school buy for the 1600 pupils?

Qn. 9. The angles of a triangle are in the ratio 1 : 3 : 5. What is the degree measure of the largest angle in the triangle?

Qn. 10. A strange cup of coffee is 97% glucose-free. How many cups of it would one have to take to get the amount of glucose in a regular coffee which is 100% caffeine?

SECTION B

Qn. 11. How many rectangles are there in the 3×3 grid made of 9 congruent squares shown below?



Qn. 12. The complete floor plan of a house is shown in the figure below. Both bedrooms have the same dimensions (in metres).

(a) What is the total area of the house?

(b) Calculate the perimeter of the house.

Qn. 13. Four years ago, David was three times as old as Joe was. In five years, David will be twice as old as Joe will. How old is David now?

Qn. 14. Carol, Jane, Kim, Nancy and Vicky, completed a 400-metre race. Nancy beat Jane by 6 seconds. Carol finished 11 seconds behind Vicky. Nancy finished 2 seconds ahead of Kim, but 3 seconds behind Vicky. By how many seconds did Kim finish the race ahead of Carol?

Qn. 15. A bus transported pupils of a certain primary school (school) from the school to Mandela National Stadium - Nambole (stadium) at a speed of 50 Km/h and returned from the stadium to school at 70 Km/h. It traveled 3 rounds within 18 hours (each round is School to stadium then stadium to school). What is the distance between the school and the stadium?

NMC 2012 PRIMARY LEVEL PAPER 1 SOLUTIONS

SECTION A : 5 marks each

1. $\frac{x}{y} = 1$ implies that $x = y$ and $x - y = 0$.

2. $-8 * 6 = -8 + \frac{6}{2} = -8 + 3 = 5$.

3. Suppose that Allen works for n weeks in total. For these n weeks she earns a total of $\$(51 + 100(n-1))$ and to get the average weekly pay, we solve

$$\frac{51 + 100(n-1)}{n} = 93$$

and this implies that $51 + 100n - 100 = 93n$. Solving this equation gives $n = 7$.

4. The sum of the angles in a polygon of n sides is given by $180(n-2)$ and for a pentagon, the sum is $180 \times 3 = 540^o$. In addition, since this is a regular pentagon, each angle is of size 108°. So, $\angle ABC = \angle BCD = 108^o$. This also means that $\angle XBC = \angle XCB = 180 - 108 = 72^o$. Therefore, $\angle BXC = 180 - 72 - 72 = 36^o$.

5. Let the numbers be $x-3, x-2, x, x+1, x+2$ and $x+3$. The sum is $7x$ and so we solve $7x = 91$ from which find that $x = 13$. The largest number is therefore $x + 3 = 13 = 3 = 16$.

6. For the Original rectangle, let the length be 6 units and the width be 2 units. The area is therefore given by $A = lw = 12$ square units. Let the width of the new rectangle be w. We know that the length of the new rectangle is 3 units. So, we solve $3w = 12$ from which we find that $w = 4$ units . The required factor is $\frac{\text{old width}}{\text{new width}} = \frac{4}{2} = 2$.

7. The area of the path is $3 \times 40 = 120$ square metres. Therefore, Kato will remain with $1600 - 120 = 1480$ square metres.

8. If x is the size of the smallest angle, then others have $3x$ and $5x$ degrees. We solve $x+3x+5x = 180$ or $9x = 180$ or $x = 20^o$. The size of the largest angle is $5x = 100^o$.

9. Each loaf serves 8 pupils (since there will be 16 slices). Therefore, 1600 pupils will be served by $\frac{1600}{8} = 200$ loaves. The school will spend Shs $200 \times 3400 =$ shs $680,000$ to feed all the 1600 pupils.

10. To get the full amount of glucose in a regular cup of coffee, you need to take $\frac{100}{3} = 33\frac{1}{3}$ cups of the strange coffee.

SECTION B

11. There are $11 \times 2 = 22$ such rectangles.

12. (a)
- Area A of one bedroom is given by $A = 20m \times 15m = 300m^2$.
- Combided area B of the kitchen and the living room is $B = 35m \times 30m = 1050m^2$.
- Total area C is given by

$$\begin{aligned} C &= A + 2B \\ &= 1050m^2 + 600m^2 = 1650m^2. \end{aligned}$$

(b) Perimeter P is given by $P = 45 + 55 + 15 + 20 + 15 + 15 + 20 = 185m$.

13. Suppose that Joe is x years now. Five years ago, she was $x-4$ years old and David was $3(x-4)$ years old. Today David is $3(x-4)+4$ years old and in five years he will be $3(x-4)+9$ years old and Joe will be $x+5$ years old. So, we solve, $3(x-4)+9 = 2(x+5$ or $3x-12+9 = 2x+10$ or $x = 13$. So, David is $3(x-4)+4 = 3(13-4)+4 = 31$ years now.

14. Let C, J, K, N and V represent the times for Carol, Jane, Kim, Nancy and Vick respectively. We are given that $N + 6 = J, V + 11 = C, N + 2 = K$ and $V + 3 = N$. Therefore,

$$K = N + 2 = V + 3 + 2 = C - 11 + 5 = C - 6.$$

So, $K + 6 = C$ implying that Kim finished 6 seconds before Carol.

15. A complete journey from School to stadium and back took $\frac{18}{3} = 6$ hours. if it takes t_1 hours to travel from school to stadium and t_2 hours from stadium to school, then, we have that

$$t_1 + t_2 = 6.$$

Since the same distance is travelled in either direction, we have that

$$50t_1 = 70t_2 \text{ which implies that } t_1 = \frac{7}{5}t_2.$$

Since $t_1 + t_2 = 6$, we have that $\frac{7}{5}t_2 + t_2 = 6$ or $\frac{12}{5}t_2 = 6$ or $t_2 = \frac{5}{2}$. Therefore, the distance from the school to the stadium is $\frac{5}{2} \times 70 = 175$km.